2.1.1 RRL

Title: Satellite-derived land surface temperature: Current status and perspectives
Author: Zhao-Liang Li , Bo-Hui Tang , Hua Wu , Huazhong Ren , Guangjian Yan , Zhengming Wan , Isabel F. Trigo , José A. Sobrino
10.1016/j.rse.2012.12.008

Physical definition of LST as radiometric skin temperature:
“How to physically interpret the results of the LST measurement remains a crucial question. As noted by Prata et al. (1995), the definition of the surface temperature may depend strongly on the type of application and the method of measurement. Because the surface temperature T_s in Eq. (4) is defined using the radiance emitted by a surface, this temperature is called the radiometric temperature (or the skin temperature) that corresponds to the radiation emitted from depths less than the penetration depth of a given wavelength (Becker & Li, 1995; Norman & Becker, 1995). The penetration depth is usually within a few millimeters in the TIR region (Wan, 1999).”

Retrieval as an "ill-posed" inverse problem requiring atmospheric and emissivity corrections:
“However, retrieving LST is still a challenging task since the LST retrieval problem is ill-posed.”
"Therefore, besides radiometric calibration and cloud screening, the determination of LSTs from space-based TIR measurements requires both emissivity and atmospheric corrections (Li & Becker, 1993; Vidal, 1991)."
"The retrieval of the LST from space is mathematically underdetermined and unsolvable (Hook et al., 1992; Kealy & Hook, 1993). The RTE described in Eqs. (3) and (4) shows that, if the radiance is measured in N channels, there will always be N + 1 unknowns, corresponding to N emissivities in each channel and an unknown LST for N equations, even if quantities other than the emissivities and LST are known a priori. Such an ill-posed problem makes the solution of the RTE sets underdetermined at ground level even if the atmospheric quantities involved in Eqs. (3) and (4) are accurately estimated. To make LST deterministic, one or more of the LSEs must be known, or the LST and LSEs have to be simultaneously solved with the aid of some assumptions or constraints on the LSEs (Dash et al., 2002; Gillespie et al., 1996; Hook et al., 1992; Kealy & Hook, 1993)."

Title: Satellite Remote Sensing of Surface Urban Heat Islands: Progress, Challenges, and Perspective
Author: Decheng Zhou ,Jingfeng Xiao ,Stefania Bonafoni ,Christian Berger,Kaveh Deilami,Yuyu Zhou ,Steve Frolking ,Rui Yao,Zhi Qiao and José A. Sobrino
https://doi.org/10.3390/rs11010048

"The surface urban heat island (SUHI), which represents the difference of land surface temperature (LST) in urban relativity to neighboring non-urban surfaces, is usually measured using satellite LST data."
"In contrast, the surface UHI (SUHI), which represents the radiative temperature difference between urban and non-urban surfaces, is primarily measured by satellite thermal remote sensing data [29]."”

There are strong biases in the study time of day and season of the selected SUHI literature. The majority of the research (69%) concentrated on the SUHI at a single time, especially the daytime (63%), while only 30% of the studies explored the SUHI during the day and night (Figure 3a). Furthermore, the data acquisition times of sun-synchronous satellites usually do not coincide with the time of day where the LST is at a minimum or maximum."
"These temporal biases of SUHI research can be attributed to data limitations, and varying magnitudes of SUHI over time. On one hand, the most widely used satellite for SUHI detection (i.e., Landsat) only has daytime data. [...] However, several studies have confirmed that SUHI intensity and the underlying mechanisms vary dramatically through time [4,6,53], which highlights the necessity of more comprehensive evaluations on SUHIs across different times of day and seasons."
"Schwarz et al. [68] provided the first comprehensive comparison analysis of indicators for quantifying the SUHI intensity and emphasized the large uncertainties associated with the quantification methods."

Title: Reexamining the relationship between surface urban heat island intensity and annual precipitation: Effects of reference rural land cover
Author: Xiaoshan Yang, Lingye Yao
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Abstract:
"Previous studies have shown that surface urban heat island intensity (SUHII) across cities is significantly positively correlated with mean annual precipitation (MAP), including linear and nonlinear correlations. Different explanations for these SUHII–MAP relations have been suggested, yet a systematic examination of the impact of variability in reference rural land cover on geographic variations of daytime SUHII and the relation with MAP is still lacking. In this study, the previously proposed SUHII–MAP relations are reexamined by investigating 60 cities across North America and 346 cities across the globe, respectively. The focus of the study is on the role of variability of reference land cover types in the SUHII–MAP relations. The 10-yr time series of satellite-observed land surface temperature, precipitation and solar radiation data, in conjunction with global land cover data, are used for analysis. Our results confirm the reproducibility of both the linear and nonlinear SUHII–MAP relations; nevertheless, the significant positive correlations between the daytime SUHII and the MAP are found to be dependent mainly on the variability of the reference land cover types associated with MAP levels (i.e., varying from desert to tree cover). In contrast, when a constant land cover type is taken as the reference, no significant correlations between the daytime SUHII and the MAP are observed. The finding highlights that respecting SUHI intensity as a relative measure and accounting for variability of reference land cover is of crucial importance for proper interpretation and understanding of SUHI."

Title: Evaluating contributions of urbanization and global climate change to urban land surface temperature change: a case study in Lagos, Nigeria
Author: Liying Guo, Lipingg Di, Chen Zhang, Li Lin, Fei Chen, Alamin Molla
https://doi.org/10.21203/rs.3.rs-1551657/v1

On the urban-rural method in Lagos, West Africa:

"Lagos metropolis as a case study is located in southwestern Nigeria, on the tropical coast of West Africa... The choice of locations for the urbanized area and its reference site is crucial to the estimations. A large Lagos area... with both urbanized areas and their suroundings at the size of $105 \text{ km} \times 73.5 \text{ km}$ is extracted... We delineate a rectangular region within the city center of Lagos totaling 100km^2 as an urban study area... where land is covered by at least 97% impervious surface... The reference site... is chosen within a non-urbanized area 30 km away from Lagos city, with the same size and shape as the urban study area. More than 95% of land in the reference site is covered by crop and little land-use conversion took place over the last twenty years. The reference site is selected based on the fact that it is in a non-urban region with a similar climate zone as urban areas but far away from the impact of urbanization on LST, and little land coversions took place over the past twenty years."

On the methodology of using this contrast:
"It can be assumed that the difference in the linear functions between the urban study area and the rural reference site can be attributed to the contribution of urbanization to ULST change. Thus, the contributions of both urbanization and global warming can be quantified by the following steps: 1) calculating annual time-series LSTs for a long period of time for both the urban study area and the reference site; 2) using linear regressions to remove the interannual variation from the time series... 3) For the urban study area, calculating the systematic trend of LST contributed by the combination of urban expansion and local impact of global warming... 4) deriving LST increase contributed by urbanization through removing the contribution of global warming from the systematic trend of LST in the reference site..."

On the ~61% metric showing localized urbanization's impact:
"Figure 5 depicts the long-term contributions of urbanization (a) and global warming (b) on the surface temperature around at four observation times. No significant difference can be observed among mean, maximum, and minimum LSTs in their contributions. On the whole, it is estimated that urbanization contributes averagely **60.97%** of the urban warming."

Title: Surface warming in global cities is substantially more rapid than in rural background areas
Author: Zihan Liu, Wenfeng Zhan, Benjamin Bechtel, James Voogt, Jiameng Lai, Tirthankar Chakraborty, Zhi-Hua Wang, Manchun Li, Fan Huang & Xuhui Lee
https://doi.org/10.1038/s43247-022-00539-x
From the Abstract:
"Warming trends in cities are influenced both by large-scale climate processes and by local-scale urbanization. However, little is known about how surface warming trends of global cities differ from those characterized by weather observations in the rural background. Here, through statistical analyses of satellite land surface temperatures (2002 to 2021), we find that the mean surface warming trend is $0.50\pm0.20~K\cdot decade^{-1}$ (mean ± one S.D.) in the urban core of 2000-plus city clusters worldwide, and is 29% greater than the trend for the rural background. On average, background climate change is the largest contributor explaining $0.30\pm0.11~K\cdot decade^{-1}$ of the urban surface warming. In city clusters in China and India, however, more than $0.23K\cdot

decade^{-1}$ of the mean trend is attributed to urban expansion. We also find evidence of urban greening in European cities, which offsets $0.13\pm0.034~K\cdot decade^{-1}$ of background surface warming."

From the Results (Urban surface warming trends):
"Urban surface warming trends. We observe faster surface warming trends at the urban core than in the adjacent rural area. The global mean surface warming trend within the urban core is 0.56 ± 0.21 K decade (mean ± one standard deviation) during the day and $0.43\pm0.16~K~decade^{-1}$ at night (Fig. 1a, b; Supplementary Table 1). Both of these trends are statistically significant $(p<0.05).$ For comparison, the surface warming trend at the rural pixels is 0.40±0.23 K decade-1 during the day and 0.37±0.21 K decade at night $(p<0.05)$ (Supplementary Fig. la, b). The daily mean surface warming trend (i.e., mean daily conditions averaged over daytime and nighttime) is 0.50± $0.20K~decade^{-1}$ at the urban core, which is 29% greater than the trend for the rural background $(0.38\pm0.21~K~decade^{-1})$. These two LST trends are both greater than the daily mean SAT trend $(0.32\pm0.083~K~decade^{-1})$ $(p<0.05)$ calculated with reanalysis SAT data for the same city clusters (Supplementary Fig. 2)..."

From the Implication Section:
"Previous urban warming studies have documented urbanization-induced regional warming by isolating the urban warming signal from in-situ SAT data or using satellite-derived LST but mostly without separating urban cores from transitional surfaces. In this study, we have divided urban pixels into urban core, rural background, and transitional areas. We demonstrate that the mean surface warming trend in the urban core of 2000-plus city clusters worldwide is 29% greater than the trend for the pixels in the rural background next to the urban land boundary. This percentage enhancement is even higher, at 56%, when referenced to the background temperature trend defined by atmospheric reanalysis data."

Title: Thermal changes along the urban-rural continuums in Southeast Asia
Author: Yuanhui Zhu*, Soe W Myint, Jiquan Chen, Peilei Fan, Karen C Seto, Atul Jain, Jiaguo Qi and Jin Wang

From the Abstract (Highlighting the outward heat bleed finding):
"This study examines LST, LCLU, and NDVI changes across 19 SEA cities of varying urbanization levels from 2000 to 2022 using Landsat images. Results show that LST change rates differ by urbanization levels, with built-up lands driving urban warming. **Larger cities exhibited more intense and widespread LST increases extending into rural areas.**"

From Section 3.2: Interdependent changes with NDVI and LCLU:
"As shown in figure 3, we analyzed the LST Sen's slope across the urban-rural gradient within a 30 km radius for 19 cities. Urban centers did not consistently exhibit the highest LST Sen's slope, indicating disparities in LST changes from urban to rural areas... Large cities generally exhibited an increase followed by a decrease in the LST Sen's slope, with turning points between 6 km and 19 km..."
"We summarized the LST Sen's slope across 19 cities along the urban-rural gradient (figure 4). For the entire study area (figure 4(a)), the LST Sen's slope peaked at the city center ($0.0303^{\circ}\text{C/year}$), decreased to 3 km, and followed a bimodal pattern with peaks at 6 km and 17 km before declining."

On the outward spread in Large Cities specifically (Section 3.2):
"Large cities (figure 4(c)) exhibited a bimodal LST Sen's slope pattern, peaking at 14 km and 17 km... The turning point occurred at 17 km, after which the slope began to decline. LST showed a decline within 4 km of the center (Sen's slope < 0), while it increased between 5 km and 26 km."

Title: Crowdsourced air temperatures contrast satellite measures of the urban heat island and its mechanisms
Author: Zander S Venter 1, Tirthankar Chakraborty 2, Xuhui Lee
DOI:10.1126/sciadv.abb9569

Discussion of the >50,000 crowdsourced stations across 342 European cities:

- **From the Abstract:**
  "Using crowdsourced citizen weather stations (>50,000) and satellite data over Europe, we estimate the CUHI and SUHI intensity in 342 urban clusters during the 2019 heat wave."

- **From the Introduction:**
  "We aimed to use the recent proliferation of crowdsourced citizen weather stations (CWSs) (28) to address gaps in our knowledge of the relative magnitude and distribution of SUHI and CUHI... We do this using CWS T_{air} measurements from >50,000 private weather stations throughout Europe during the month of July 2019."

Exact statistic on the sixfold overestimate (1.45°C vs 0.26°C):

- **From the Abstract:**
  "Satellites produce a sixfold overestimate of UHI relative to station measurements (mean SUHI $1.45^{\circ}C;$ CUHI $0.26^{\circ}C$), with SUHI exceeding CUHI in 96% of cities during daytime and in 80% at night."

- **From the Results (CUHI versus SUHI):**
  "We found that city-scale UHI was overestimated by $1.19\pm0.02^{\circ}C$ (mean ± SE, $n=342$) if LST was used instead of Tair (mean SUHI $1.45^{\circ}C$ versus mean CUHI $0.26^{\circ}C$; Figs. 1 and 2A). For the cities considered during July 2019, this reflects a sixfold overestimation in the urban contribution to local temperature if satellite observations are used instead of ambient air temperature measurements."

Title: Integrating Satellite-Based and Participatory Spatial Analyses for Local, Contextualised, and Nuanced Extreme Heat Hotspot Mapping in Overheated Urban Southeast Asia
Author: Laurence L. Delina, Homer Pagkalinawan, Sharon Feliza Ann Macagba
https://doi.org/10.1002/sd.70329

"Crucially, participatory mapping supplied fine-scale, lived-experience data that

(a) identified microhotspots that LST products missed, (b) revealed exposure pathways linked to travel distances and service access, and (c) highlighted social vulnerabilities and coping practises that inform equitable intervention priorities."

"Participatory mapping—where residents identify and describe places of heightened heat exposure and associated coping strategies—offers a complementary lens that captures local knowledge, behavioural adaptations, and place-based vulnerability that are invisible to satellite imagery (Guardaro et al. 2022; Lehnert et al. 2023)."

"City-level analyses, therefore, often fail to resolve block-scale variability or capture where and how residents actually experience heat in daily life... Beyond instrument readings, participatory methods document where people seek shade, avoid outdoor activities, or experience heat-related symptoms; they reveal socially salient hotspots—narrow alleys, markets, schoolyards—that satellite products may not prioritise."

"Participatory mapping and citizen science provide a complementary source of fine-grained, context-rich information. When residents map where they feel heat, seek shade, alter activities, or experience heat-related symptoms, they reveal behavioural patterns, infrastructure gaps, and socially salient hotspots—such as market stalls, narrow alleys, or schoolyards that remote sensing alone may not prioritise..."